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MILITARY SPECIFICATION

MODIFICATION KIT, BILATERAL RANGING, FOR RADIO SET AN/ARN-21B

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall, when integrated with Radio Set AN/ARN-21B in a military aircraft, provide a bilateral ranging function against a similarly equipped aircraft. The addition of the bilateral ranging function shall in no way degrade the performance of Radio Set AN/ARN-21B when used in its normal TACAN mode of operation.

1.2 Classification - The Bilateral Ranging Modification Kits shall be of two types consisting of the following items:

1.2.1 Modification Kit R58-1 ARN21-C-1

Items	Appl. Para.
Modulator, Solid State Air-to-Air	3.4.1
Modification Parts, Receiver - Transmitter RT-220B/ARN-21	3.4.2
Modification Parts, Mounting MT-928/ARN-21	3.4.3
Modification Parts, Radio Set Control C-1763/ARN-21A	3.4.4

1.2.2 Modification Kit R58-2 ARN21-C-1

Items	Appl. Para.
Modulator, Solid State Air-to-Air	3.4.1
Modification Parts, Receiver-Transmitter RT-220B/ARN-21	3.4.2
Modification Parts, Mounting MT-928/ARN-21	3.4.3
Modification Parts, Radio Set Control C-866/ARN-21	3.4.5

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-R-7060	Radio Set, AN/ARN-21(*)
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-C-18308	Control, Radio Set C-866/ARN-21
MIL-C-18640	Control, Radio Set C-1763/ARN-21A

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Ave., Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Parts and material requirements shall conform to Specification MIL-E-5400.

(b) Component parts differing from those employed in previously produced models shall not be used without the approval of the procuring activity.

3.1.1 Non-standard Parts and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron Tubes shall not be used. Transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682 except that all transistors and diodes must be of the silicon type.

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3.2 Design and Construction - This equipment shall conform with all applicable requirements of Specification MIL-E-5400 for design, construction, and workmanship, unless otherwise specified herein.

3.2.1 Operating Life - The equipment shall have a minimum total operating life of 3000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.2.2 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.3 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment:

Temperature	Room Ambinet 25°C + 15° C -5°C
Altitude	Normal Ground
Vibrations	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 VAC and 27.5 ± 0.5 VDC

3.2.4 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein.

3.2.5 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves II and IV of Specification MIL-E-5400.

3.2.6 Power Requirements - The power requirements for the solid state air-to-air modulator shall not exceed the following under any combination of normal operating conditions:

0.5	Amperes	@ 124 volts RMS 320 to 1000 cps
0.25	Amperes	@ 29 Vdc
0.16	Amperes	@ 120 Vdc
1.5	MA	@ -100 or -180 Vdc

3.2.7 Installation - The nature of the modifications and additions to the Radio Set AN/ARN-21B shall be such as to permit field modification by normally assigned maintenance personnel. No cutting, drilling, or reworking of fabricated parts shall be allowed.

3.3 Performance - When the equipment covered by this specification is properly installed in the Radio Set AN/ARN-21B (Specification MIL-R-7060), three modes of operation shall be provided as follows (the modified Radio Set AN/ARN-21B nomenclature is Radio Set AN/ARN-21D):

(1) Receiver (REC) - In this mode the Radio Set AN/ARN-21D shall perform all the functions and meet all the requirements of Radio Set AN/ARN-21B operating in the receive mode.

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(2) Transmit-Receive (T/R) - In this mode the Radio Set AN/ARN-21D shall perform all the functions and meet all the requirements of Radio Set AN/ARN-21B operating in the transmit-receive mode.

(3) Air-to-Air (A/A) - In this mode, two Radio Sets AN/ARN-21D forming a bilateral link shall simultaneously give a continuous display of the link distance when operating in compatible air-to-air frequency channels. In addition, it shall be possible for at least five aircraft with this equipment, under conditions specified herein, to simultaneously obtain and display their distance from a sixth similarly equipped aircraft. The sixth aircraft shall display its distance from one of the other five.

3.3.1 Operational Range (A/A) - The operational range shall be from zero out to a line of sight distance of 195 nautical miles.

3.3.2 Accuracy - The equipment shall have a distance accuracy in the air-to-air mode of ± (0.1 mile plus 0.2% of the distance) for signal strengths giving satisfactory performance in the transmit-receive mode. The tracking error in the air-to-air mode shall be no worse than specified by Specification MIL-R-7060 for Radio Set AN/ARN-21B at 85% reply efficiency.

3.3.3 Tracking Rate (A/A) - The equipment shall be capable of locking on and continuously tracking distance replies while moving at speeds relative to the other end of the link up to the basic equipment capability in the normal mode when operating with a beacon which is 85% efficient in replying to distance interrogations.

3.3.4 Sensitivity (A/A)

3.3.4.1 Lock-On - The range lock-on sensitivity in the air-to-air mode shall be no less than the range lock-on sensitivity in the transmit-receive mode at 70% reply efficiency.

3.3.4.2 Transponding - The interrogation level required to effect 70% transponding efficiency in the air-to-air mode shall be no greater than 6db above the 70% lock-on sensitivity in the normal (T/R) mode.

3.3.5 Traffic Handling Capability - It shall be possible to transpond to as many as five interrogating Radio Set AN/ARN-21D in the air-to-air mode with not less than 88% reply efficiency provided the interrogation signal strengths do not differ from each other by an amount determined by the air-to-air AGC adjustment of the transponding equipment.

3.3.6 Automatic Gain Control (A/A) - An adjustable air-to-air AGC shall be provided such that it is possible to reject the weaker of two signals when they differ by any amount between 15 and 55db. This AGC shall be generated by properly coded interrogations and shall require no more than 2 seconds to reach 90% of the final value determined by any instantaneous change in the interrogation signal level.

3.3.7 Range Speed of Response (A/A) - A complete range search cycle shall not exceed 30 seconds.

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3.3.8 Range Memory (A/A) - Loss of the distance reply signal shall not cause the system to go into search for 10 ± 2 seconds when previously tracking for at least 1 second.

3.3.9 Range Display (A/A) - The air-to-air distance shall be delayed in a manner identical to the normal TACAN mode of operation using the same indicators. The displayed reading shall be retained during the memory interval.

3.3.10 Transmitter Power (A/A) - When the total transmission rate is between 22 and 30 pulse pairs per second, the peak power shall be the same as that transmitted in the normal mode when tracking a distance reply signal from a ground beacon. A maximum reduction in peak power of 3db is permissible when the total transmission rate is 1050 pulses per second.

3.3.11 Amplitude Modulation (A/A) - When transmitting in the air-to-air mode, the maximum noise amplitude modulation shall not exceed 2db.

3.3.12 Modulation Characteristics (A/A) - The interrogation pulses shall have characteristics identical with those in normal TACAN operation. In addition, each equipment shall transpond to each pair of received interrogations with a single pulse having the same charactertistics as the interrogation pulses.

3.3.13 Transmitter Muting - Satisfactory muting of the transmitter shall be provided during channel selection in the air-to-air mode and in the T/R mode, as well as while switching from A/A to T/R or from T/R to A/A.

3.3.14 Over-Interrogation - The transmitter shall not be capable of transponding at a rate in excess of 1250 pulses per second.

3.3.15 System Delay (A/A) - The transponder reply pulse shall occur 62 ± 1.0 microseconds after receipt, at the antenna terminal, of the first pulse of a coded interrogation pair. The distance indicator of the interrogating Radio Set AN/ARN-21D shall display 0 ± 0.1 miles upon receipt of a reply pulse at its antenna terminal 62 microseconds after transmission of the first interrogation pulse.

3.4 Detail Requirements

3.4.1 Modulator, Solid State Air-to-Air - The modulator shall employ two separate sequentially triggered line type pulse forming circuits, using solid state discharge switches, for the generation of the coded interrogation pulses. The interrogation rate shall be the same in both the transmit-receive (T/R) and air-to-air (A/A) modes of operation, and will conform to the requirements of Specification MIL-R-7060. The Solid State Air-to-Air Modulator shall replace the AN/ARN-21B tube modulator on a directly interchangeable basis. Normal service adjustments are permitted.

3.4.1.1 Functions - In addition to performing all functions performed by the Radio Set AN/ARN-21B Radio Modulator, the Solid State Air-to-Air Modulator shall contain the air-to-air decoding, delay, and transponding circuits necessary for the bilateral ranging mode of operation. Solid State circuitry shall be employed throughout. Time sharing of the first interrogation pulse

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generator shall be employed for the generation of the air-to-air reply pulses. A dead time gate shall be provided to prevent triggering during the recovery period. Specifically, the Solid State air-to-air Modulator shall contain the following circuits:

(1) Interrogation PRF Generator
(2) Interrogation Encoder
(3) Transmission Pulse Shaping
(4) A/A Decoder
(5) A/A System Delay
(6) A/A AGC
(7) A/A Over-Interrogation Countdown
(8) Transmitter Muting
(9) First Pulse Generator Dead Time

3.4.1.2 Weight - The weight of the solid state air-to-air modulator shall not exceed 50 ounces.

3.4.1.3 Dead Time Gate - All time shared random triggers, both interrogation and reply, shall be inhibited for 125 ± 25 microseconds after the generation of either the first interrogation pulse or an air-to-air reply pulse.

3.4.1.4 Adjustments - Screwdriver adjustments shall be provided to set the interrogation PRF, the interrogation pulse pair spacing, the air-to-air system delay time, and the air-to-air AGC.

3.4.1.5 Suppression Pulse - The Air-to-Air Solid State Modulator shall include the necessary circuits for generating a suppression pulse output in accordance with Specification MIL-R-7060 for air-to-air reply pulses as well as the interrogations.

3.4.1.6 Interrogation Sync - An interrogation sync terminal shall be provided on the modulator chassis. This terminal shall be readily accessible when the receiver-transmitter dust cover is removed for maintenance or test.

3.4.2 Modification Parts RT-220B/ARN-21 - Shall consist of wires, cables, connectors, terminals, nameplates, and hardware necessary to convert the Receiver-Transmitter RT-220B/ARN-21 into an RT-220D/ARN-21 when the Solid State Air-to-Air Modulator is installed. No change in inter-connecting cables or aircraft wiring shall be permitted. The modification shall be such that attempted operation with the AN/ARN-21B tube modulator shall not cause damage to either the modulator or the RT-220D/ARN-21. The RT-220D/ARN-21 shall be interchangeable with the RT-220B/ARN-21 for normal TACAN operation with any combination of modified or existing AN/ARN-21B components. The modification kit shall provide the Receiver-Transmitter RT-220D/ARN-21 with the capability of receiving on any one of 126 crystal controlled image frequencies when the air-to-air mode of operation is selected.

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3.4.3 Modification Parts, MT-928/ARN-21 - Shall consist of a prewired relay and bracket assembly necessary to add the bilateral ranging function selection capability to the shock mount. A nameplate shall be included for identifying the modified Shock Mount, MT-928A/ARN-21. The MT-928A/ARN-21 shall be interchangeable with the MT-928/ARN-21 for normal TACAN operation with any combination of modified or existing AN/ARN-21B components.

3.4.4 Modification Parts, C-1763/ARN-21A - Shall consist of a new mode selection switch and front panel for adding the air-to-air mode selection capability to Radio Set AN/ARN-21D. A nameplate shall be included for identifying the modified Radio Set Control, C-1763A/ARN-21A. The C-1763A/ARN-21A shall be interchangeable with the C-1763/ARN-21A for normal TACAN operation with any combination of modified or existing AN/ARN-21B components. When placed in the A/A Mode, the C-1763A/ARN-21A shall effect the necessary switching to place the complete Radio Set AN/ARN-21D in the A/A Mode. When used with an MT-928A/ARN-21 and an RT-220B/ARN-21, the C-1763A/ARN-21 placed in the A/A Mode shall result in the normal T/R Mode of system operation. When used with an MT-928/ARN-21, the C-1763A/ARN-21 placed in the A/A Mode shall remove all primary power from the receiver-transmitter.

3.4.5 Modification Parts, C-866/ARN-21 - Shall consist of a switch, potentiometer, panel, wires, and necessary hardware for interchanging the physical configuration of the mode selector switch and identification volume control. In addition, the air-to-air mode selection capability shall be added, and its functional operation shall be identical to the C-1763A/ARN-21A. A nameplate shall be provided for identifying the modified Radio Set Control, C-866A/ARN-21.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements. Functional tests of the equipment covered by this specification are to be conducted after installation in Radio Set AN/ARN-21B which will be provided as Government Furnished Equipment. It is recognized that failures which may occur will not necessarily be chargeable to the equipment covered by this specification, but rather a fault in the basic equipment being used as a test vehicle. Nevertheless, since it is a requirement of this specification that the addition of the bilateral ranging function shall in no way degrade the performance of Radio Set AN/ARN-21B when used in the normal TACAN mode of operation it shall be the responsibility of the contractor to analyze each failure and demonstrate to the satisfaction of the procuring activity that each failure encountered is in no way attributable to the equipment covered by this specification:

(1) Preproduction Tests

(2) Production Sample Tests

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(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial test laboratory.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. This data may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contracotr's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

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4.2.5 Production Equipment - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection or if already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Preproduction Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

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4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Tests - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Tests - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration	Any selected frequency within the range of 20 to 30cps (excluding resonant points) and a minimum amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall mechanically cycled periodically through the various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 200	1
	1 for each additional 500 or fraction thereof

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4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing reliability test specified in para. 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipments.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.
(2) On an early equipment after an engineering change
(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests
(2) Vibration Test (60 minutes cycling in each of 3 planes)
(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

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(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure (s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test, then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 3500 or fraction thereof

4.5.1 Test Conditions - The life testing to be conducted shall apply to Item 1, the Solid State Air-to-Air Modulator. It shall be acceptable to perform life test operation using an abbreviated or simulated RT-220B environment wherein appropriate input signals and power are provided and appropriate artificial loads are employed. Provision shall be made to measure performance characteristics at specified intervals either in the simulated environment or in an Air-to-Air AN/ARN-21B maintained for that purpose.

Temperature	Normal Room
Altitude	Normal Ground (0 - 5000 ft.)
Humidity	Room Ambient
Primary A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
Primary D.C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

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4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Natureof the failure
(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Use - The equipment covered by this specification is intended for use in field modification of Radio Set AN/ARN-21B, to provide Air-to-Air Ranging operation in military aircraft.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages 27.5 ± 0.5v DC and 115 ± 1.0v AC, as applicable. These data are to be used in testing the equipment at installation points for compliance with the minimum acceptable standard of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specification - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.5 Ordering Data - Procurement documents should specify the following:

(1) Title, number and data of this specification.

(2) List of items of 1.2 comprising required kits.

(3) Selection of the applicable levels of preservation, packaging and packing.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/8 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 48-R300

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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Chief, Bureau of Naval Weapons (RREN-5)

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